

Progress Report #1

Project Title: Email Spam Detector: Chrome Extension
with Python Backend

Course: 4495 – Applied Research Project, Section: 002

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Work Logs

Date	Hours Worked	Description of Work Done
Jan 14, 2025	2	Conducted initial research on spam detection techniques, including keyword filtering, heuristic algorithms, Bayesian classifiers, and machine learning. Compiled findings into a research document
Jan 16, 2025	1.5	Researched tools and components required for Chrome extension development. Explored Chrome API documentation and feasibility of integrating with Gmail.
Jan 20, 2025	3	Defined project structure, created a preliminary framework for backend and frontend integration, and discussed Flask API implementation. Documented key design decisions.
Jan 22, 2025	0.5	Created project repository, added initial files and organized folder structure for efficient project management. Encountered difficulty in designing an optimal file structure for seamless integration between frontend and backend.
Jan 25, 2025	1.5	Finalized project proposal, incorporated feedback, and documented technical specifications. Submitted proposal for review.
Jan 27, 2025	1.5	Research done on the dataset for project.
Jan 30, 2025	2	Started working on project using Jupyter, with the help of Python.
Feb 01, 2025	1.5	Faced difficulty in pushing folders on GitHub, tried to figure out the problem.
Feb 05, 2025	2.5	Worked together as team to find out errors encountered while data cleaning
Feb 07, 2025	2	Worked on pushing files using command prompt and visual studio.
Feb 08, 2025	3	Created project report, analysed the work done and challenges faced in project.
Feb 09, 2025	2.5	Finalised the project report and submitted for grading.

Description of Work Done

This week, I conducted extensive research on spam detection methodologies, focusing on their applications in a browser-based email filtering system. The research included a comparison of traditional methods (keyword filtering, heuristic analysis, Bayesian classifiers) and more advanced approaches (machine learning, NLP-based filtering). I explored the feasibility of integrating these techniques into a Chrome extension powered by a Python Flask backend. Additionally, I structured the project repository, organizing folders for backend and frontend components. The proposal was completed and submitted, outlining technical requirements, project goals, and development strategy.

One of the major challenges faced was designing the file structure in a way that includes pushing the created folder using the GitHub. Initially, the integration was problematic due to differences in how browser scripts and server-side applications handle requests. This was mitigated by restructuring the folder hierarchy and clearly defining API endpoints to facilitate smooth data transfer.

Challenges also included finding an optimal method for real-time email extraction within Gmail while maintaining privacy and security. We are trying to mitigate it by researching Chrome API permissions and reviewing Gmail add-on development guidelines.

Repo Check-in of Implementation Completed

The following files/folders have been checked into the repository:

- **/README.md**: Describes the project and its folder structure.
- **/ReportsAndDocuments/**: Contains the initial research document detailing spam detection methodologies and the finalized project proposal.
- **/Implementation/**: Includes the initial Flask backend structure and placeholder files for Chrome extension integration.
- **/repo_setup/**: Added ReadMe documentation and project organization details.
- **/Misc/**: Contains additional reference materials and supporting files for team collaboration.

Work is progressing as planned, and the next focus will be implementing Flask API endpoints and building a preliminary spam detection model.

Next Steps:

- Develop an initial version of the Chrome extension with email data extraction functionality.
 - Implement a basic spam detection mechanism using keyword filtering.
 - Conduct preliminary testing to evaluate accuracy and efficiency.
 - Regularly update the repository with incremental developments.
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Conclusion

In this phase of the project, I have successfully conducted research on spam detection techniques, set up the project repository, and finalized the project proposal. Despite challenges in file structuring and integration between frontend and backend, solutions were implemented to streamline communication between components. The groundwork has been laid for the next phase, where the focus will be on developing core functionalities, testing, and refining the spam detection algorithm. Continuous learning and iteration will be key to ensuring the project meets its objectives effectively.